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E-241 BUILDING AND EQUIPMENT GROUNDING AND LIGHTNING PROTECTION

1. SCOPE OF WORK

The work shall include the design, supply, manufacturing, factory testing, delivery to the site, installation, commissioning and testing of grounding and/or lightning protection of all equipment and structures for the electrical installation and/or wiring in accordance with Part 4, Subsection 1.2 Summary of Work.

The grounding and lightning protection system shall include, but not be limited to, the followings:

- (1) Connecting structural support steel of the buildings and structures to ground riser cables at the support steel foundations.
- (2) The proper installation and bonding of cable trays, conduits and copper ground conductors to the grounded steel structures to establish an indoor grounding system.
- (3) Grounding of all electrical equipment and enclosures and accessories to the grounded steel structures or by interconnecting to the indoor grounding system.
- (4) Lightning protection for all the structures listed in Part 4, Subsection 1.2 in accordance with the criteria specified in Clause 4.5 herein.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Subsection 2.5 Codes and Standards and the following specific standards:

2.1 National

- PUIL : Peraturan Umum Instalasi Listrik - Indonesia
(General Regulation of Electrical Installation
Indonesia).
- SPLN : Standard Perusahaan Umum Listrik Negara.

2.2 International Electrotechnical Commission

- IEC 60228 : Conductors of insulated cables.

IEC 60364 : Electrical Installation of buildings.

2.3 American National Standards Institute/Institute of Electrical and Electronics Engineers

ANSI/IEEE 142 : Grounding of Industrial and Commercial Power System.

2.4 National Electrical Manufacturers Association

NEMA CC-1 : Electric Power Connectors for Substations.

2.5 National Fire Protection Association

NFPA 780 : Lightning Protection

3. DESIGN REQUIREMENTS

All ground conductors shall be soft drawn copper with Class 2 stranding as defined by IEC 60228.

The minimum main ground conductor and safety ground conductor sizes for motors and equipment with the same kW rating shall have a cross-sectional area at least equivalent to:

- That of the live conductor for cross-sectional areas less than 25 mm²
- 25 mm² for cross-sectional area between 25 mm² and 50 mm²
- 50 % of that of the live conductor for cross-sectional areas exceeding 50 mm².

The grounding terminal shall be identified in accordance with IEC 60445.

Switchgear and 400 Volt motor control center (MCC) ground buses shall be grounded by the Contractor using two (2) conductors, one (1) at each end of the switchgear internal ground bus. The minimum conductor sizes shall be as shown below.

<u>Equipment Fed From</u>	<u>RMS Symmetrical Fault Current Level</u>	<u>Cable Size</u>
6 .3kV Switchgear	50 kA	300 mm ²
400 V Switchgear	50 kA	300 mm ²
Motor Control Centers	50 kA	300 mm ²

The Contractor shall protect all joints between dissimilar metals against galvanic action.

Ground connection shall be made by exothermic welding or bolting. In general, ground connections at equipment shall be by bolting.

Ground risers from the station buried grounding system shall be provided by Contractor at or near building and structure foundation. The connection of ground risers to building steel and structures shall be by the Contractor. The connections shall be made by exothermic welding.

The Contractor shall design and install system of supplementary copper ground conductors in large building with no structural steel. The supplementary ground conductors shall be 6 x 50 mm copper bars or 300 mm² cable. The supplemental ground conductors shall be banded to the reinforcing steel at each floor and shall be interconnected so that there are at least two independent paths to the outdoors grounding system from any piece of equipment.

4. INSTALLATION REQUIREMENTS

4.1 General

In general grounding shall be designed and installed in accordance with IEEE 142 or IEC and as specified herein.

Conductive materials enclosing electric conductors or equipment, or forming part of such equipment shall be grounded to prevent local potential differences on these materials, which may be hazardous to personnel and to provide a low impedance fault path to facilitate overcurrent device operation in case of insulation failure or ground faults.

AC power circuits shall be grounded at the source to limit voltages from lightning, line surges, unintentional contact with higher voltage circuits, faults and to limit the voltage to ground to a safe value during normal operation and to facilitate overcurrent device operation.

The indoor grounding system shall consist of a network of cable trays, conduits, and copper ground conductors which are connected together and to the steel structure of the building. The indoor grounding system shall be connected at

several locations, by the Contractor, to the outdoor grounding system (a network of interconnected buried cables and/or driven ground electrodes around the site including underneath and/or perimeter of buildings, structures and other metallic objects) through connections between the buried outdoor grounding system and building steel. Connection shall be 6 x 50 mm² copper bar or 300 mm² copper cable. Equipment frames, equipment neutrals, metal structures, metallic armor of cables, cable sheaths, and cable sheathing tapes shall be connected to the grounding system.

Bonding jumpers shall be used, as required, to ensure electrical continuity. Also supplementary bare copper grounding conductors shall be used to increase conductivity and current carrying capacity at locations where the conduit system, etc. provides an inadequate path for ground fault current.

4.2 Cable Trays

Cable trays shall be connected to grounded building steel with copper bonding jumpers of at least 300 mm² at thirty (30) meter intervals or in at least two places for shorter runs. The bonding jumpers may be omitted where cable tray supports are welded to the trays and the supports are welded to building steel.

Copper bonding jumpers, having an area of at least 300 mm², shall be used at expansion joints or sections of the cable tray system where required to ensure that an electrically continuous path to the power source neutral and a substantial metal ground is established. Electrical enclosures and junction boxes which are bolted or welded to cable trays are considered adequately grounded.

4.3 Conduits and Cable Armor

Most of the ground fault current will flow in conduits and cable armor even when ground conductors are provided between equipment and the nearest ground bus or building steel. Therefore, all joints and connections in the conduit and cable tray systems shall be clean and tight to provide an adequate, electrically continuous path.

A separate ground wire need not be provided in the conduits or the metallic armor of cables, if the ground connections are clean and tight and if the conductivity is such that it fully meets the following requirements:

- (1) The maximum temperature rise due to a single line-to-ground or double line-to-ground fault shall not exceed 250°C.

- (2) The conductance is equal to or greater than that of the phase conductors.
- (3) If the above requirements are not met, a copper ground conductor of the size required in Section 3. herein shall be run with the power conductors.

All flexible conduits and liquid tight flexible metal conduits used for power cables shall be provided with bonding jumpers. The bonding jumpers shall have conductivity equal to that of the phase conductor, but in no case shall the jumpers be smaller than 16 mm².

A bonding jumper shall be used when a conduit is terminated at a cable tray but is not directly connected to the cable tray by welding or an approved grounding cable tray clamp.

Suitable grounding jumpers shall be installed at all junction boxes, pull boxes, expansion fittings, and at flexible conduit sections in the conduit run to establish adequate ground continuity in the conduit system.

All hangers and metallic junction boxes for any non-metallic conduit runs shall be connected to the cable tray system or to building steel.

Cable armoring may be used for grounding electrical equipment subject to review by the Owner.

4.4 Grounding Connections

4.4.1 Motors

Motors frames shall be connected to building steel at the nearest practical location by a bare copper safety ground conductor.

Small motor frames (< 5 kW) shall be grounded through the conduit to the cable tray system.

4.4.2 Transformers

Indoor power transformer neutrals shall be connected to the nearest building steel or similar ground connection.

Enclosures for small transformers rated 400 volts and 5 kVA and below may be connected to the cable tray system by the conduit of the cable supplying the

equipment. This conduit shall be connected to the cable tray system and to the transformer enclosure. A copper safety ground shall be provided if armor is used as a ground conductor in lieu of conduit.

4.4.3 Electrical Equipment

All electrical equipment provided with ground bus bars such as switchgear, motor control centers, distribution panels and control cubicles, etc., shall be provided with grounding conductors bolted to both ends of the ground bus bars and connected to the nearest building steel.

Indoor surge arrester neutrals shall be connected to the frame of the protected equipment and ground by a direct metallic path.

4.4.4 Cable Sheaths and Shielding Tapes

The insulation shield for control cable and metallic armor for three-phase power cables and multi-conductor power cables shall be grounded at both ends.

Rubber or synthetic insulated cables with a shielding tape, but without magnetic sheath or armor, shall be grounded with 10 mm² or larger tinned copper wire.

4.4.5 Lighting

Where electric metallic tubing (EMT) is used for installation of indoor 220 volt AC lighting, ground conductors shall be installed along with the power supply cables.

Indoor and outdoor lighting where rigid steel conduits are used for installation, are considered grounded, provided that the conduit system is properly installed to provide electrical continuity. Bonding jumpers shall be used at cable trays and discontinuities in the conduit system such as sliding pull boxes, etc.

Lighting power distribution cabinet enclosures shall be grounded to building steel or the cable tray system. Enclosures shall not be grounded by connection to the lighting system neutral feed wire. Lighting neutrals shall be insulated from enclosures and grounded at power distribution center transformer neutrals only.

4.4.6 Miscellaneous Equipment

Bonding jumpers and/or grounding conductors (as part of control cable) shall

be used, as required, for control stations and other accessories, which are not in direct contact with grounded steel structures or may not be grounded through conduit systems.

The neutral of instrument transformers and/or relays shall be grounded only at one (1) location, as close to the transformers or relays as possible. Neutral return conductors shall be in the same cable as the phase conductors.

All crane rails shall be connected to building steel in at least two places.

The ground conductor of a control circuit shall be in the same cable that contains the other conductors in the circuit. If the control circuit occupies more than one (1) cable, a ground conductor shall be included in each of the cables.

4.5 Lightning Protection

4.5.1 General

The Contractor shall provide lightning protection systems for all buildings and structures based on the following criteria:

The lightning protection system shall be designed and installed in accordance with NFPA 780, except air terminal locations, which shall be determined based on the Striking Distance Method.

4.5.2 Air Terminals

All air terminals shall be 600 mm long of stainless steel. Air terminals shall be located so that an imaginary sphere 35 meters in radius will not be able to touch any part of the building or structure but will rest instead on the air terminals and on the ground (thirty five (35) meters striking distance). Grounded metal objects 5 mm or more in thickness do not require air terminals but must be grounded. Such objects may be considered as serving the function of air terminals. However, air terminals shall be provided as needed to protect thin walled miscellaneous accessories, which are not self protected.

4.5.3 Down Conductors

At least two independent down conductors connected to the ground rod in grounding mat shall be provided on each structure. Additional down conductors shall be provided, if the perimeter of the structure exceeds seventy five (75) meters, so that an additional down conductor is provided for each 30 meters of

perimeter or fraction thereof.

All down conductors shall be of copper with a cross sectional area of at least 120 mm². The conductors shall be supported and protected in accordance with NFPA 780. Any splices in down conductors shall be made by exothermic welding, bolting or with substantial compression sleeves of the type. Joints embedded in concrete or otherwise concealed shall be by exothermic welding. Compression sleeves are preferred to bolting when their use is appropriate.

Structural steel building members shall not be used as down conductors. However, the requirements for bonding to structural steel in NFPA 780 shall be strictly adhered to.

4.5.4 Reinforcing Bars

The Contractor shall ensure that the structure reinforcing bars in any concrete work are electrically continuous from the top to the bottom of the structure. Isolated sections of reinforcing are not acceptable.

The normal overlapping and wiring together of the reinforcing is considered adequate for electrical continuity, provided that each successive level of rebar is wired to the lower level at each junction.

As soon as erection of the structure reinforcing is started, the Contractor shall make permanent electrical grounding connections to ground riser furnished in place next to the structure foundations. This shall consist of the installation of 6 x 50 mm copper penetration bars, 300 mm above grade level. Each bar shall be bonded to all layers of rebars it crosses. The penetration bars shall be installed at locations corresponding to the down conductors of the permanent lightning protection system.

Anchor studs of embedment plates shall be electrically bonded to the rebars. Lap binding wire can be used for this purpose.

Rebars shall be extended 600 mm above top of the exposed outdoor structure(s) at equally spaced locations. These rebars will act as temporary air terminals for the protection of the structure until the permanent air terminals have been installed.

Before the roof slab is poured, 120 mm² copper conductor pigtail connections shall be made to steel rebars at locations corresponding to the permanent location of the down conductors.

4.5.5 Permanent Lightning Protection

The down conductors for each concrete structure shall be connected to the pigtails at the top and to the embedded copper bars at the bottom of the structure.

A 120 mm² copper conductor ring shall be installed at the top of each structure, interconnecting the air terminals.

All equipment and accessories and all metal objects at the top of a structure shall be connected or bonded to the lightning protection system.

All electrical and electronic equipment and accessories which may be exposed to surges such as, outdoor control panels, devices and outdoor high intensity arc discharge lighting fixtures, shall be provided with lightning surge arresters. The surge arresters shall be of the metal oxide type suitable for a low voltage power supply system. Electronic type varistors shall not be used as a substitute.

The top and bottom of each ladder-way shall be connected to the lightning protection system.

All pipes or conduits running outside the structure wall shall be adequately grounded. They shall be bonded to the lightning protection system as required by NFPA 780.

4.5.6 Additional requirements for Bypass and HRSG Stack Lightning Protection

- (1) Each metal chimney may be used for air terminals in accordance with clause 4.5.2 requirements.
- (2) Foundation Grounding

The Contractor shall ensure that the bypass stack and HRSG foundation reinforcing and the steel pile jackets are interconnected to form an electrically continuous structure.

At each down conductor location, the Contractor shall install a 6 x 50 mm copper bar in the foundation. Each copper bar shall be connected to each layer of reinforcing steel, preferably by welding at the non load bearing ends of the reinforcing. The copper bar shall be welded to the steel jacket of the nearest pile.

All steel pile jackets shall be interconnected by welding to 300 mm² copper cable or 6 x 50 mm copper bars.

The copper bars shall be exothermically welded to embedded plates for connecting each stack grounding system to the outdoor grounding system. The plates shall be drilled and tapped with a NEMA 4-hole pattern as described in NEMA CC-1.

All bare copper installed below grade shall be embedded in concrete.

(3) Permanent Lightning Protection for Stacks

A set of 120 mm² copper down conductors shall be installed at the locations called for in Clause 4.5.3. The top sixteen (16) meters of the down conductors and jumpers, connectors, etc. within sixteen (16) meters of the top of each stack shall be lead covered as called for by NFPA 780. The down conductors shall be connected to each of the copper bars listed in Section 4.5.4. Paragraph 3.

Each down conductor shall be connected to the stack at the top by the most direct route feasible using flexible jumpers.

Exterior platforms shall be connected to the down conductors. Appropriate measures shall be taken to prevent corrosion between dissimilar metals.

If ladders are not welded to the platform they shall be electrically connected to the platform or a down conductor at both the top and bottom of the ladder. All intermediate breaks in the ladders shall be electrically bonded to provide a continuous metallic circuit from top to bottom.

All pipes and conduits shall be bonded to the lightning protection system at the top and bottom of their run.

5. INSPECTION AND CERTIFICATION OF INSTALLATION

Should the local Authority or the Owner's insurance require special inspection, testing and certification, the Owner will advise the Contractor of such requirements, notwithstanding the conditions of Part 2. This Work shall be included in the Contract.

6. QUALITY ASSURANCE/QUALITY CONTROL

Quality assurance/quality control for building and equipment grounding and lightning protection shall conform to the requirements included in Part 4. Section 2, Subsection 2.7.

The quality assurance/quality control documentation as specified shall include, but not limited to, the followings:

- (1) Installation Inspection Report.
- (2) Installation Test Reports.
- (3) All documentation required by applicable codes and standards.

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